

VARIABILITY OF PRECIPITATION IN THE ATACAMA DESERT: ITS CAUSES AND HYDROLOGICAL IMPACT

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ABSTRACT

An analysis of the variability of rainfall at 27 stations and run-off at 4 stations between 18° and 28°S in the Atacama Desert has been carried out. A diagonal boundary zone between summer- and winter-dominated areas is related to the provenance of the rainfall: Amazonia to the north and east, and Pacific moisture to the south. It is shown that winter rainfall tends to be higher during El Niño years, while heavy summer rainfall tends to be more common during La Niña. However, rather than the precipitation being directly controlled by El Niño–Southern Oscillation (ENSO), previous studies have shown that it is the regional synoptic conditions towards the source areas that largely control temporal precipitation variations, and these are in turn either facilitated or inhibited by ENSO. The spatio-temporal variability of precipitation leads to a complex hydrological regime. Perennial rivers in the north and central Atacama Desert tend to flood in summer, especially during La Niña conditions, from source to sea. Perennial rivers in the south tend to flood in summer, but as a result of melt from the previous years snowfall, especially during El Niño conditions, again from source to sea. However, while inland areas may also experience flooding of ephemeral rivers in summer associated with La Niña, coastal areas on the other hand experience winter flooding of ephemeral rivers associated with El Niño. Surface water flood events, and groundwater recharge events reported in the literature, are generally less frequent than ENSO events, confirming the requirement for specific synoptic conditions and making the use of averages unsound for present-day hydrological studies and water resource evaluations. Copyright © 2006 Royal Meteorological Society.

KEY WORDS: precipitation; run-off; hyper-aridity; ENSO; Atacama Desert; Chile

1. INTRODUCTION

In arid zones especially, climate has the most important control on hydrological processes. Whereas topography and geology may control the location and timing of surface water run-off or groundwater recharge, meteorology determines whether there will be any run-off or recharge in the first place. It is therefore of primary importance to investigate and evaluate the climate for a full understanding of the hydrological system.

Recent studies on the tropical climatology of South America (e.g. Garreaud and Aceituno, 2001; Markgraf, 2001), and especially the Altiplano (Garreaud *et al.*, 2003 and references therein), have contributed greatly to a deeper understanding of precipitation variability and its causes.

The climate of the Atacama Desert is largely controlled by two factors: firstly, its zonal location between 15° and 30°S (Figure 1) in the sub-tropical high-pressure belt where descending stable air produced by the Hadley circulation significantly reduces convection and hence precipitation; and secondly, the upwelling cold Peruvian Current that inhibits the moisture capacity of onshore winds by creating a persistent inversion that traps any Pacific moisture below 1000 m above sea level (a.s.l.). Additionally, the proximity of the Andean Cordillera upwind restricts moisture advection from the east and a largely decoupled boundary layer

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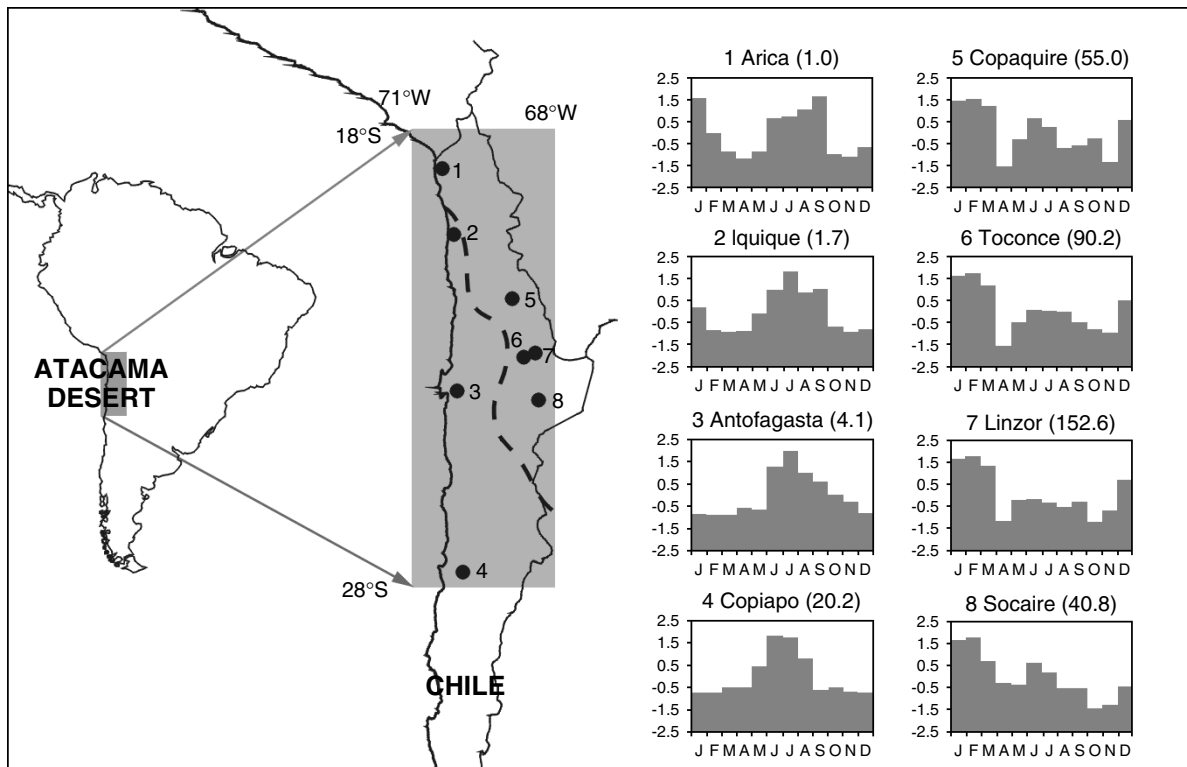


Figure 1. Location map of northern Chile with monthly frequency plots of rainfall for four long-term coastal stations with dominant winter rainfall and four short-term Andean stations with dominant summer rainfall. The dividing line between stations with peak rainfall in summer or winter is shown dashed. Each station is standardized using an LN3 transformation. Mean annual rainfall (mm) given in brackets

circulation cell above the inversion, caused by insolation effects over the Western Cordillera and Altiplano, leads to subsidence return flow over the Central Valley. This 'Rutllant cell' is considered to be instrumental in generating hyper-aridity (Rutllant *et al.*, 2003).

During the austral summer (DJF) in the Atacama Desert, wet episodes tend to occur throughout the Western Cordillera and Altiplano when strong upper level easterly winds enhance moisture transport from Amazonia creating saturation during uplift within deep convection cells (Garreaud *et al.*, 2003). As a consequence of the easterly moisture source, a rain shadow develops over the Western Cordillera and Atacama Desert, and mean annual precipitation declines rapidly from over 300 mm year⁻¹ at 5000 m a.s.l. to less than 20 mm year⁻¹ at 2300 m a.s.l. (Houston and Hartley, 2003). Below 2300 m a.s.l., associated with the Central Valley, is a zone of extreme hyper-aridity in which the mean annual precipitation is less than 1 mm year⁻¹. Winter rainfall is largely sourced from northerly and easterly moving frontal systems originating from the Pacific (Vuille and Ammann, 1997), and within the core area of the Atacama contributes less than 30% of the mean annual rainfall.

The Atacama Desert thus straddles the boundary between two climate zones: to the north lies the tropical summer rainfall zone and to the south lies the mid-latitude winter rainfall zone. Although the climate is hyper-arid in the core zone between 15° and 30°S and between sea level and 3500 m, a few perennial rivers, such as the Lluta, Loa and Copiapo, cross the desert sourced exogenously and by drainage from aquifers recharged in the Andean Cordillera and Pre-Cordillera.

The impact of El Niño on the western coast of South America has long been known, and recent studies (e.g. Diaz and Markgraf, 2000) have extended this to its causal mechanisms and global impacts. Higher-than-average precipitation associated both with El Niño (Ortlieb, 2000; Vargas *et al.*, 2000) and its contrary

state, La Niña (Vuille, 1999; Garreaud and Aceituno, 2001), is seasonally and spatially dependent and has important hydrological consequences.

Here, using historical observational data, the temporal and spatial variability of precipitation in Northern Chile is examined and the contribution of ENSO to this variability is investigated. On the basis of flow gauging data for several rivers, together with a number of flood studies, we show that these variations in precipitation have direct consequences for the hydrology of the region, resulting in a complex system that can, at least partly, be understood as a response to the alternating ENSO states.

2. DATA AVAILABILITY

The database used to evaluate northern Chilean rainfall patterns is based on historical data recorded by the Dirección Meteorología and the Dirección General de Aguas (Table I). Several long-term stations date from the mid-nineteenth century with ten times as many stations since the mid-twentieth century, spread over the western slopes of the Andes between 19° and 28°S from sea level to 4200 m a.s.l.

Long-term data between 1870 and 2000 are available for four coastal stations, although the data prior to 1900 are considered somewhat unreliable. Data for the period 1977–2000 are available for 23 additional

Table I. Rain gauge stations used in the analysis. Mean value given for water-year, November–October

Station	Long. S (Decimal degrees)	Lat. W (Decimal degrees)	Elevation (m) a.s.l.	24-Year mean annual precipitation	Fraction in summer (NDJFMA)
Antofagasta ^a	23.45	70.45	10	3.1	0.14
Arica ^a	18.50	70.27	5	0.9	0.49
Ascotan	21.68	68.28	3956	71.4	0.90
Ayquina	22.28	68.32	3031	28.4	0.89
Calama	22.47	68.92	2260	4.2	0.34
Camigna	19.32	69.42	2380	13.4	0.95
Caspana	22.32	68.22	3260	63.3	0.90
Chiu Chiu	22.33	68.65	2524	4.4	0.65
Copacquire	20.95	68.90	3490	54.1	0.87
Copiapo ^a	27.35	70.21	380	22.5	0.13
Coya Sur	22.45	69.65	1290	0.4	0.00
Coyacagua	20.02	68.82	3990	120.1	0.93
El Tatio	22.37	68.03	4320	153.9	0.90
Guatacondo	20.93	69.05	2460	11.5	0.86
Inacaliri	22.03	68.07	4100	129.5	0.95
Iquique ^a	20.22	70.13	10	1.5	0.27
Linzor	22.20	67.98	4096	154.6	0.93
Ollague	21.22	68.25	3650	78.1	0.95
Parca	20.02	69.20	2570	27.3	0.91
Peine	23.68	68.07	2480	18.1	0.74
Potrerrillos	26.40	69.47	2850	22.1	0.29
Pumire	19.13	69.12	4200	140.3	0.97
Quillagua	21.63	69.52	802	0.15	0.00
Sagasca	20.18	69.33	1815	0.9	0.84
Socaire	23.58	67.88	3350	40.7	0.78
Toconce	22.25	68.17	3350	94.1	0.92
Ujina	20.97	68.65	4200	151.3	0.90

^a Data for 1900–2000.

stations with a good geographical spread and means close to the long-term average, including both wet years (1984–1986 and 1997) and dry years (1978–1979 and 1988–1990).

Standard rain gauges in use at these stations probably reflect only precipitation in the form of rainfall. Both snowfall at high elevations (Vuille and Ammann, 1997; Vuille and Baumgartner, 1998) and fog at low elevations (Aravena *et al.*, 1989) are under-recorded. Furthermore, the gauges are located in sites with different exposure and aspect. All the longer-term stations have changed their location and, in some cases, the type of recording instrument over the period. Nevertheless, it is considered that the processed data (see the following text) represent a reliable record of the rainfall variability in the region.

Monthly data was checked for missing or incomplete values. Stations with more than 2% missing monthly data were excluded from the analysis. The remaining missing data were estimated by inserting the mean monthly value, factored by the annual rainfall for the year compared with the long-term station mean. Specific annual outliers were investigated for anomalous monthly values and corrected in the same way as the missing data. Correction of data amounted to less than 1% of all months for those stations used in the analysis. Monthly values were converted to water years (November–October). Since the annual data are positively skewed with a zero lower bound, the annual time series for each station was standardized assuming a three-parameter log-normal (LN3) distribution. Furthermore, since station means vary over 3 orders of magnitude, standardization facilitates inter-station comparison. None of the stations used in the final dataset show any significant trend towards wetter or drier conditions over the period of record.

Monthly mean run-off data for four rivers, based on mean daily flow data from hydrographs, were obtained from the Dirección General de Aguas (Table II). The length of record varies from 11 to 22 years. Between 21 and 32% of the daily data are incomplete, but since flows are serially correlated it is possible to estimate months with missing or no data using linear interpolation without significant loss of accuracy. Monthly values were converted to water years (November–October). The effect of abstractions and impounding reservoirs means that some modifications to the natural flow regime occur, and absolute values are not directly comparable. Nevertheless, by standardizing each station over its period of record, again using an LN3 transformation, it is possible to compare their generalized flow characteristics.

ENSO data were sourced from the Climate Research Unit at the University of East Anglia; the southern oscillation index (SOI) is based on Ropelewski and Jones (1987) and Allan *et al.* (1991). Sea-surface temperature anomalies (SSTA) were sourced from NOAA-CPC (<http://www.cpc.ncep.noaa.gov/data/indices/index.html>). Monthly values of SOI and SSTA were converted to water years (November–October) to be comparable with the precipitation and flow data. El Niño and La Niña events are taken from Quinn *et al.* (1987), Quinn (1992) and Ortlieb (2000) prior to 1950, and from NOAA-CPC (http://www.cpc.noaa.gov/products/analysis_monitoring/ensostuff/ensoyears.html) after 1950.

3. SEASONAL AND SPATIAL VARIATIONS

As will be shown, it is not possible to consider spatial variations of precipitation in the Atacama Desert without also taking into account seasonal patterns, since they are intimately linked as a result of the provenance of the precipitation.

Table II. Flow gauge stations used in the analysis. Mean annual flow (MAF) values are given for water-year, November–October

Station	Long. S (Decimal degrees)	Lat. W (Decimal degrees)	Elevation (m) a.s.l.	Catchment (km ²)	Start year	MAF (m ³ s ⁻¹)	Specific discharge (l s ⁻¹ km ⁻²)
Lluta	18.40	70.30	10	3447	1986–	1.33	0.39
Loa	21.42	70.07	0	32 820	1990–	0.26	0.01
Salado	22.28	68.32	3031	770	1979–	12.27	15.93
Copipo	27.80	70.18	758	8343	1984–	1.86	0.22

Monthly frequency plots (Figure 1) show a coastal and southern zone dominated by austral winter (JJAS) rainfall. Despite the annual decrease from 20 mm year⁻¹ at 27°S to 1 mm year⁻¹ at 18°S, a summer component (JF) becomes increasingly important north of 22°S, approaching 30% of annual rainfall at Arica. The northern Andean zone, by contrast, is dominated by summer (DJFM) rainfall, although winter precipitation still occurs throughout.

The distribution of summer and winter rainfalls is shown in Figure 2. Winter rainfall shows strong latitudinal control with a tenfold increase for every 5° of latitude south of 26°S (Figure 2). Summer rainfall amounts are considerably greater than the winter ones (Figure 2), but decrease rapidly with declining elevation as a result of the rain shadow effect created by the Andes (Houston and Hartley, 2003). Between 18° and 24°S the relationship between mean annual rainfall (*MAR*, mm yr⁻¹) and elevation (*A*, m a.s.l.) is best described by the exponential relationship (Figure 3):

$$MAR = e^{0.0012A} \quad (r = 0.94, \quad p < 0.01) \quad (1)$$

The division between these two zones occurs at the limit of influence of the two sources of precipitation and creates a dry diagonal extending from Chaca (18.82°S, 70.14°W, 145 m a.s.l.) in the northwest, where the *MAR* is 0.1 mm yr⁻¹ (DGA, 1987) to Paso San Francisco (at ca 27.0°S, 67.7°W, 3700 m a.s.l.) in the Andean Cordillera to the southeast where the *MAR* is ca 30 mm yr⁻¹. Associated with this boundary, especially in the Central Valley between 19° and 25°S at around 1000 m a.s.l., is a zone of extreme hyper-aridity.

4. INTER-ANNUAL VARIATIONS

4.1. Winter coastal rainfall

The winter-rainfall-dominated coastal zone shows considerable variation over the last hundred years but with no significant trend (Figure 4). Wet years were noticeable during the late 1920s, around 1940 and 1959, and again in the late 1980s and 1990s. Wet years greater than 1 standard deviation (σ) have a recurrence

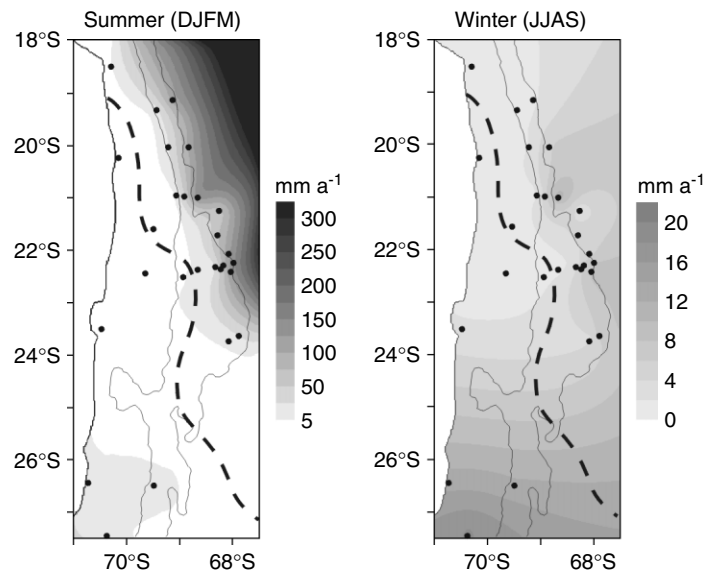


Figure 2. Mean annual rainfall for the period 1977–2000 summer and winter months in the central Atacama Desert showing elevational (rain shadow) control in the north and east, and latitudinal control in the south. Station data is contoured using a kriging algorithm (Cressie, 1991). Note the difference in scales for summer and winter, with most of the precipitation falling in summer. The dry diagonal and associated zone of extreme hyper-aridity can be clearly seen. Topographic contours are shown for 2000 m and 4000 m a.s.l.

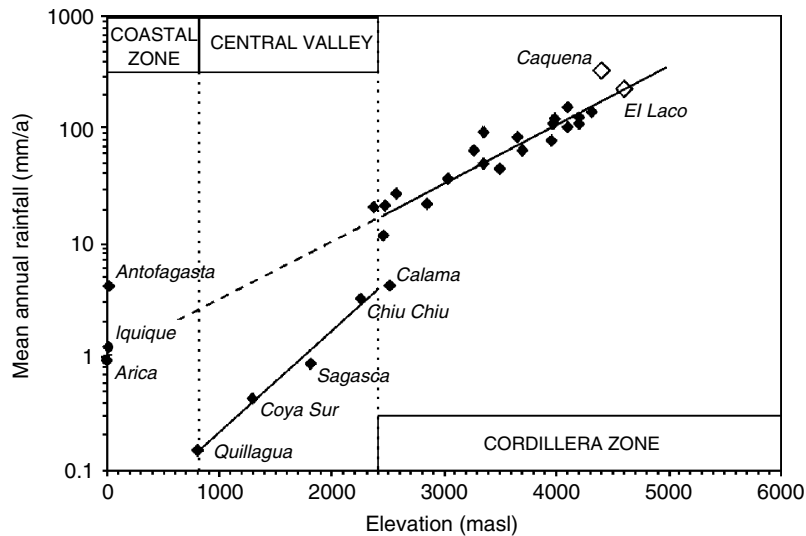


Figure 3. Precipitation–elevation relationships for the Atacama Desert between 18° and 24°S. An exponential decline with decreasing elevation is due to the impact of the Andes on northeasterly airflows from Amazonia creating a rain shadow. The zone of extreme hyper-aridity is associated with the Central Valley and the boundary between summer and winter precipitation zones. Single point data for El Laco (Vuille, 1996) and Caquena (Fuenzalida and Rutllant, 1986) have been added

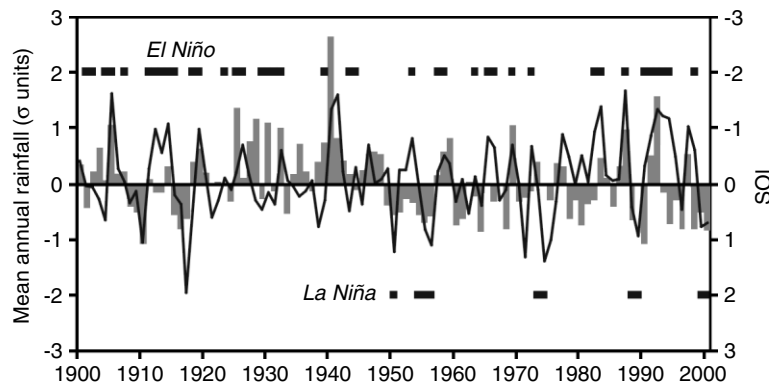


Figure 4. Time series of coastal stations compared with the SOI (reversed axis) and ENSO events. The histogram is based on the mean annual standardized rainfall for Arica, Iquique, Antofagasta and Copiapo using an LN3 transformation

interval of 11 years and are all associated with negative values of the SOI and positive SSTA for the Pacific Ocean Niño region 3 (5°N–5°S, 150°–90°W).

Mean annual rainfall along the coast shows a significant correlation with SOI and SSTA throughout the last 100 years (Table III), although only 16–17% of the variance in precipitation is accounted for by ENSO, and this is largely due to the winter component.

On the other hand, *ca* 50% of years when El Niño conditions prevailed (based arbitrarily on SOI < −1, SSTA > +1) resulted in wet years, and in particular, the negative SOI years of 1914 and 1983 failed to produce wet conditions in the coastal Atacama Desert. By contrast, in Santiago at 33°S (and Copiapo) both the 1914 and 1983 El Niño conditions did give heavy rainfall and 87% of El Niño years resulted in wet conditions (Rutllant and Fuenzalida, 1991).

Table III. Correlation matrix between mean standardized precipitation and ENSO parameters. Correlations significant at 95% are in bold, 99% underlined and 99.9% asterixed

	SOI	SSTA Niño 3
<i>1900–2000 (coastal stations)</i>		
Mean annual precipitation	<u>−0.401</u>*	<u>0.481</u>*
Summer precipitation	−0.169	0.246
Winter precipitation	<u>−0.319</u>	<u>0.454</u>*
<i>1977–2000 (non-coastal stations)</i>		
Mean annual precipitation	0.012	0.040
Summer precipitation	0.186	−0.234
Winter precipitation	<u>−0.516</u>	<u>0.568</u>
<i>1977–2000 (non-coastal stations)</i>		
Annual maximum daily rainfall	0.317	<u>−0.476</u>

4.2. Summer Andean rainfall

The summer-rainfall-dominated Andean zone has a greater number of data stations within the study area, but over a shorter period of time. Variations in winter (MJJASO) and summer (NDJFMA) rainfall for non-coastal stations are shown in Figure 5. Significant winter rainfall ($>1\sigma$) has a recurrence interval of six years during the 24-year period and its coincidence with El Niño is evident (see also Table III), although only 25–27% of the variance in winter precipitation is explained by ENSO.

Variations in summer precipitation tend to show a reversed relationship with ENSO, being generally higher during La Niña conditions (1984, 1999, 2000), although the wet summer of 1987 and to a lesser extent 1997 were associated with the development of El Niño conditions. The correlation between ENSO and wet summers is not significant (Table III); La Niña explaining less than 3% of the variance. However, a plot of the annual daily maximum rainfall at the same stations (Figure 6) shows a closer correspondence with La Niña conditions. The relationship is significant at 95% (Table III), explaining 10–18% of the variance.

4.3. Frequency

Recurrence intervals for coastal wet winter conditions is 12 years for 1σ and more than 100 years for 2σ . Recurrence intervals for both winter and summer wet conditions away from the coast are 6–8 years for 1σ . By comparison based on the analyses of Quinn *et al.* (1987), Quinn (1992) and Ortlieb (2000) for the period 1900–1950 and NOAA-CPC since 1950, ENSO events (both El Niño and La Niña) have had a recurrence interval of 3.3 years, rather more frequent than heavy rainfall in either the coastal or Andean zones.

Typical mean annual rainfall frequency curves for coastal and Andean Cordillera stations, together with annual daily maximum rainfall at Andean stations, are shown in Figure 7. Coastal stations show a pattern of increasing rainfall from north to south as expected owing to the latitudinal control over stations with winter-dominant rainfall. Andean stations show greater complexity, however; Linzor and Toconce are stations in the upper Turi Basin which is backed by an amphitheater of volcanic cones that exert local topographic (dynamic) control on airflows, generating increased rainfall in both amount and intensity (Houston, 2006). By comparison, Copaquire, Socaire and Ujina at similar elevations on the Andean Cordillera to the north and south (Table I) of the Turi Basin have considerably lower precipitation amounts and intensity.

5. RUN-OFF

5.1. Flow regimes

The annual flow regime and location of four perennial rivers that cross the Atacama Desert are shown in Figure 8. The rivers are all ultimately sourced in the Andes but receive gains from groundwater drainage

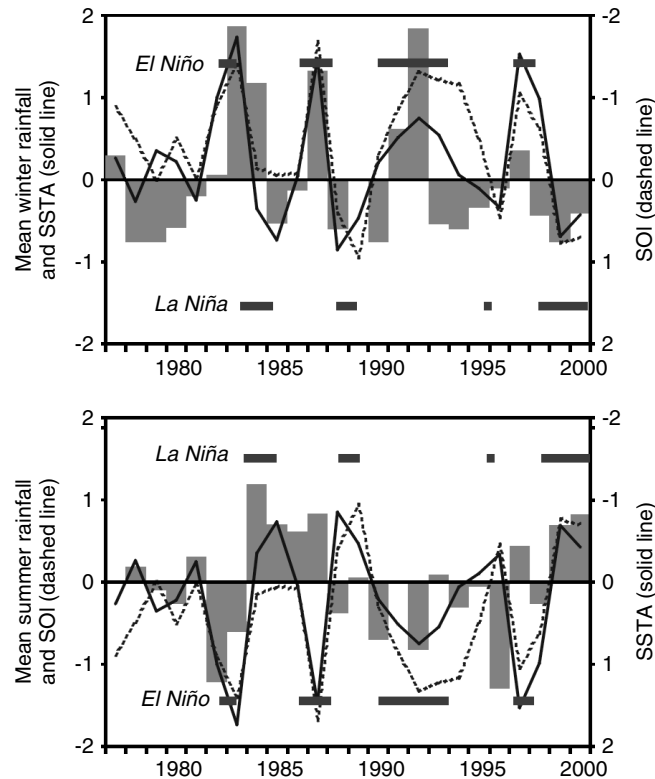


Figure 5. Time series of winter (MJJASO) and summer (NDJFMA) rainfall for non-coastal stations compared with the SOI and SSTA in the region El Niño 3, and ENSO events. The histograms are mean annual standardized station data transformed using an LN3 distribution. Note the reversals of ENSO data and axes to show the relationship between winter rainfall and El Niño, and summer rainfall and La Niña

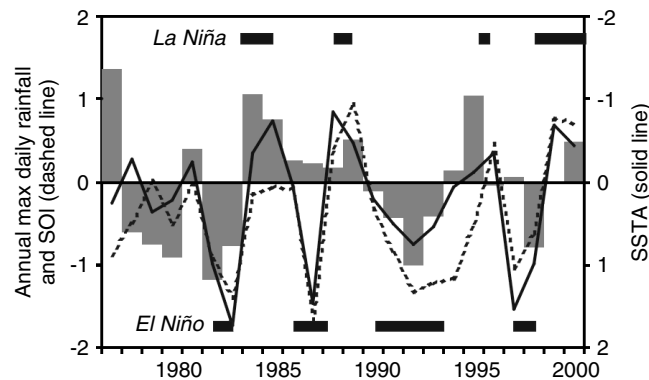


Figure 6. Time series of annual maximum daily rainfall (standardized using an LN3 transformation and averaged for all non-coastal stations) showing the relationship with La Niña. Note reversals of ENSO data and axes to show the relationship between annual maximum daily rainfall and La Niña

and losses due to evapotranspiration along various sections of their courses, buffering their response to precipitation, and this needs to be taken into consideration during analysis.

The Río Lluta flows from east to west, consequent upon the western slope of the Andes. Its flow regime is dominated by peak run-off during summer (DJFM) with recession during the rest of the year. This response is undoubtedly due to run-off from summer precipitation at higher elevations.

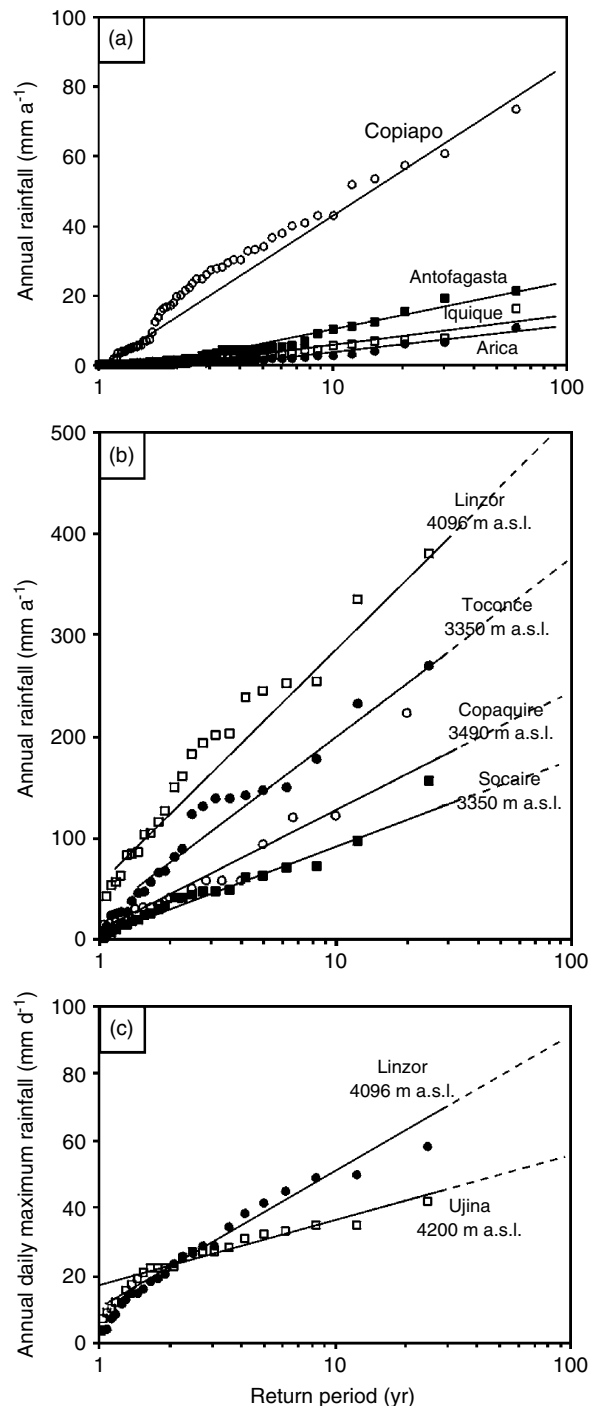


Figure 7. Frequency of annual rainfall for (a) coastal and (b) Andean stations. Frequency of annual daily maximum precipitation for Andean stations (c). Note the different rainfall scales

The Río Loa is the only river that has significant north–south reaches likely due to geological controls, which add greatly to its catchment area and allows extensive hydraulic contact with several aquifers (Houston, 2006). There are two clearly defined peak flow periods: February, and late winter (ASO). The February peak

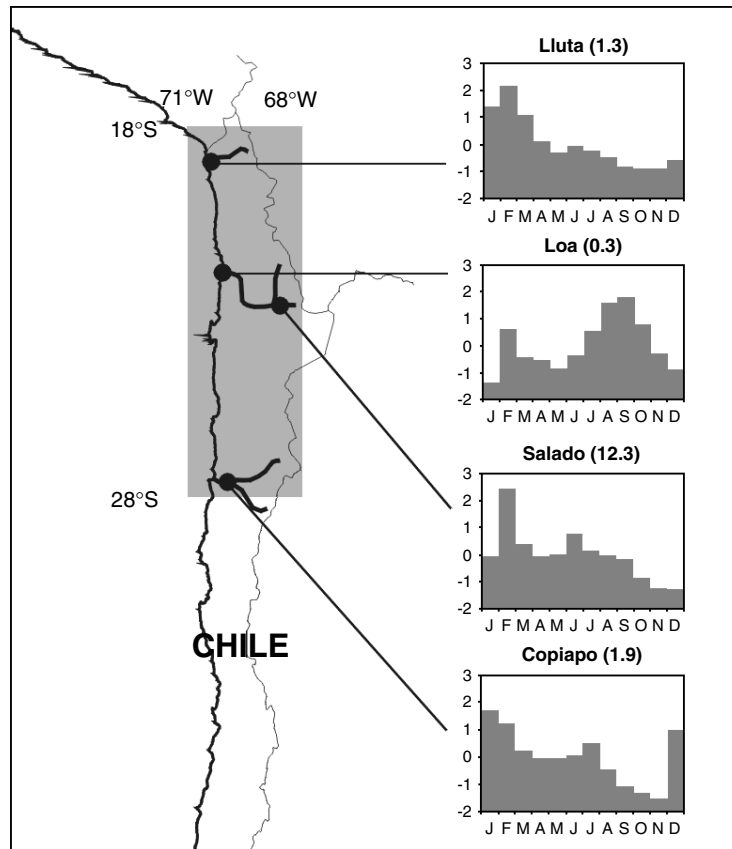


Figure 8. Location map of northern Chile with monthly frequency plots of run-off for four perennial rivers. Each station is standardized using an LN3 transformation. Mean annual flow ($\text{m}^3 \text{s}^{-1}$) over 11–22 years given in brackets (see also Table II)

is a response to summer precipitation, but the late winter peak shows greater amplitude and period than the summer peak and does not display the characteristic hydrograph shape of a fast rise and slow fall. This peak is unlikely due to winter rainfall in the Andes, since winter rainfall is less than 10 mm at this latitude. The most likely explanation for this flow is groundwater discharge as a result of (a) summer rainfall recharge, lagged due to the buffering effect of storage in the aquifers, and (b) decreased catchment evaporation losses during the winter. The relatively low flow rate of the Río Loa is partly due to a large percentage of the catchment area being at low elevations and hence receiving little rainfall, partly due to its location within the dry diagonal, and partly due to significant abstraction within the catchment.

The Río Salado is a tributary of the Loa, located in the Andes with peak flows characteristically in February due to summer rainfall in the Andes, but there is a small subsidiary peak in winter (June), which might be due to either winter precipitation, or more likely, lagged groundwater drainage buffered by aquifer storage. The relatively high flow rate of the Río Salado is largely due to its catchment location at higher elevations on the western slope of the Andes, which maximizes run-off and minimizes losses due to infiltration, evaporation and abstraction relative to the Río Loa.

The Río Copiapo is a consequent draining from east to west. It has a flow regime similar to the Río Salado, with a primary peak in summer (DJF) and a secondary peak in winter (July). Summer rainfall at this latitude is very low, whereas the winter rainfall is higher (Figure 2), and it is likely that considerable winter precipitation in the form of snow goes unrecorded. Hence, the main summer peak flow is due to snowmelt (in common with all rivers further south, see for example Waylen and Caviedes, 1990), while the smaller winter peak might be due to direct winter rainfall or lagged groundwater drainage.

Table IV. Correlation matrix between river flows, precipitation and ENSO parameters. Correlations significant at 95% are in bold and at 99% underlined

		Rainfall		SOI	SSTA Niño 3
		Summer	Winter		
Lluta	1986–2000	<u>0.701</u>	−0.362	0.315	−0.193
Loa	1991–2000	<u>0.587</u>	−0.108	0.616	−0.369
Salado	1979–2000	0.600	−0.205	0.463	−0.331
Copiapo	1984–2000	−0.065	−0.085	0.332	−0.032
Copiapo + 1 year		<u>0.578</u>	0.452	−0.110	0.229

5.2. Inter-annual flow variations

Table IV confirms that there is a significant relationship between annual run-off and summer rainfall for all catchments apart from the Copiapo, which is significant at lag1 for both summer and winter rainfalls. This confirms that peak flow in the Río Copiapo is a result of the spring-melt of the preceding year's summer and winter precipitation that would have been largely in the form of snow. This also accounts for the slightly earlier (DJ) occurrence of the summer peak in the Río Copiapo compared with the other rivers (JF) seen in Figure 8. Hence the time series shown in Figure 9 includes the Río Copiapo advanced by one year and compared with winter rainfall. Not unexpectedly, there is an overwhelming control of mean annual flow by annual precipitation, which explains 69% of the variance after 1984, when there is data for more than one station.

Figure 10 and Table IV show the relationship between mean annual run-off and the ENSO parameters. The three rivers that show significant correlations with summer rainfall also show a correlation with La Niña, whereas the Río Copiapo (which shows a significant correlation with rainfall at year 1 lag) is correlated with El Niño (note the sign reversal), although not significantly so.

5.3. Frequency

Figure 11 shows the frequency plots for the four gauging stations. Although the flow records are relatively short, the recurrence interval for flows greater than 1σ is between 5 and 11 years, similar to the recurrence intervals of wet years, and less frequent than the ENSO recurrence intervals. This has been confirmed by flood analysis in the Central Valley (Houston, 2002) and Calama Basin (Houston, 2006), where significant floods have been found to occur on decadal or centennial scales.

6. DISCUSSION

6.1. ENSO impacts

The extent to which ENSO controls the variability of precipitation and hence the hydroclimatology of the Atacama Desert is not yet fully clear and is the subject of continuing debate (Ortlieb, 2000; Dettinger *et al.*, 2000; Garreaud *et al.*, 2003; Vuille and Keimig, 2004). A superposed event analysis for three El Niño and three La Niña years (Figure 12) suggests that considerable ENSO control is exerted over both winter and summer precipitation. The analysis is based on the average anomalies (σ at each station assuming an LN3 distribution) for the three years indicated for summer and winter precipitation, contoured using a kriging algorithm (Cressie, 1991).

Positive precipitation anomalies associated with El Niño are confined to the coast during summer but extend throughout the central and southern Atacama during the winter. Drought conditions (negative precipitation anomalies) are associated with El Niño in the Altiplano during both summer and winter confirming previous studies (Vuille, 1999; Vuille *et al.*, 2000; Garreaud and Aceituno, 2001). The fact that only 50% of El

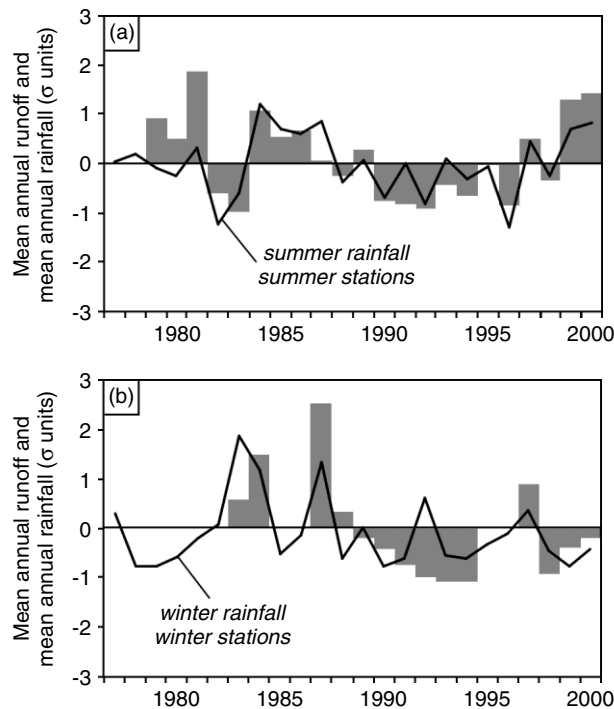


Figure 9. The relationship between flow and rainfall. (a) Time series of mean annual flow for the Ríos Lluta, Loa and Salado compared with mean annual summer rainfall of summer stations given in Table I. Prior to 1984 flow data is for one station only. (b) Time series of annual flow in Río Copiapo advanced by one year compared with mean annual winter rainfall of winter stations given in Table I. All data standardized using an LN3 transformation for each station

Niño years lead to wet winters along the Atacama coast compared with 87% further south at Santiago reinforces the concept of latitudinal control and suggests that special conditions are required to generate a response further north. Rutllant and Fuenzalida (1991) and Montecinos and Aceituno (2002) showed that the synoptic conditions associated with El Niño which lead to increased winter precipitation are related to the occurrence of a blocking high in the Belingshausen Sea (50°S , 90°W) which forces westerly storm tracks northwards. Variations in the strength and position of the blocking high are therefore likely to control the extent to which moisture penetrates northwards as far as the Atacama Desert and account for the generally low variance explained by ENSO on wet conditions in the coastal Atacama Desert despite its underlying control.

Positive precipitation anomalies associated with La Niña are largely confined to the Andean Cordillera and Altiplano during summer, with near neutral conditions during winter throughout the Atacama. Several previous studies have also shown the correspondence between wet summers in the Altiplano and La Niña (e.g. Vuille, 1999; Garreaud and Aceituno, 2001). However, the variability in summer rainfall explained by ENSO is low on a seasonal basis, but increases for daily maximum amounts. Garreaud *et al.* (2003) showed that the synoptic conditions that generate intense summer rainfall are related to convective activity over the Altiplano, which is strengthened during periods of enhanced easterly airflow with adequate moisture transport from Amazonia. Such atmospheric circulation conditions are largely forced by tropical Pacific sea-surface temperatures (SSTs), hence the link with ENSO, but greatly depend on the zonal positioning of the anomalous easterly airflow (Vuille and Keimig, 2004), thereby creating significant variability in the linkage.

It is therefore possible to infer that while ENSO may facilitate precipitation variability in the Atacama Desert, it does not directly drive it; synoptic conditions that are favoured by ENSO but not wholly controlled by it lead to precipitation extremes. This is underlined by the frequency of wet years, which have return periods of between 6 and 12 years compared with 3.3 years for ENSO events.

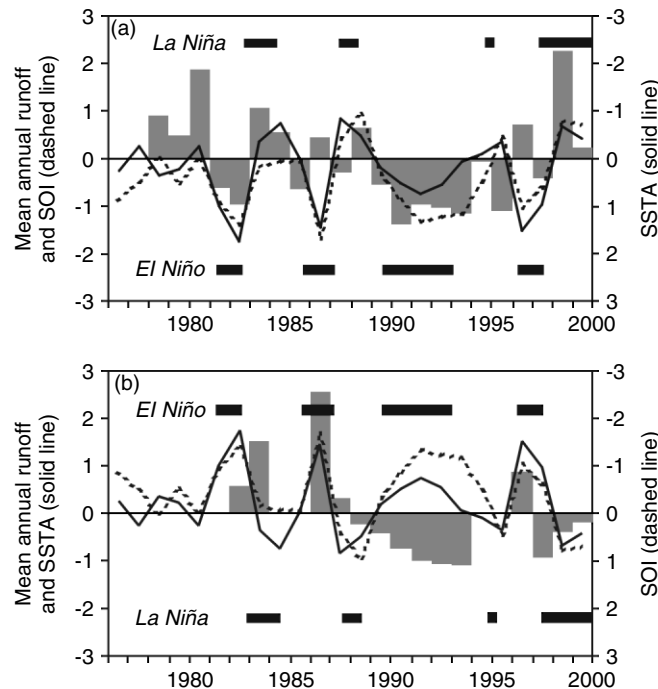


Figure 10. Time series of Ríos Salado (a) and Copiapo + 1 (b) annual run-off (standardized using an LN3 transformation) compared with the SOI and SSTA in region El Niño 1 + 2 and ENSO events. Note the reversals of ENSO data and axes to show the weak relationship between Río Salado run-off and La Niña, and Río Copiapo and El Niño

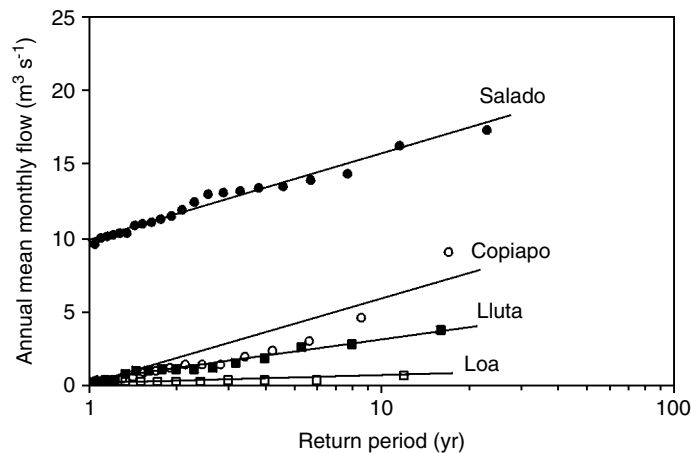


Figure 11. Frequency of mean annual run-off for the perennial rivers in northern Chile

So how far does ENSO control hydrological events such as surface water floods and groundwater recharge? Several studies have shown a link between streamflow and ENSO (e.g. Quinn, 1992; Garreaud and Rutllant, 1996; Ortlieb, 2000; Dettinger *et al.*, 2000; Dettinger and Diaz, 2000), but until recently relatively few show a link with groundwater recharge (e.g. Houston, 2002, 2006). Where a linkage has been found between the ENSO-driven precipitation and run-off, the non-linear nature of the hydrological response means that the latter is greatly magnified (Dettinger *et al.*, 2000; Houston, 2006). Despite a consensus view that hydrological events are forced by ENSO, there remain some misconceptions, particularly that the coastal flooding in the Atacama Desert is driven only by El Niño.

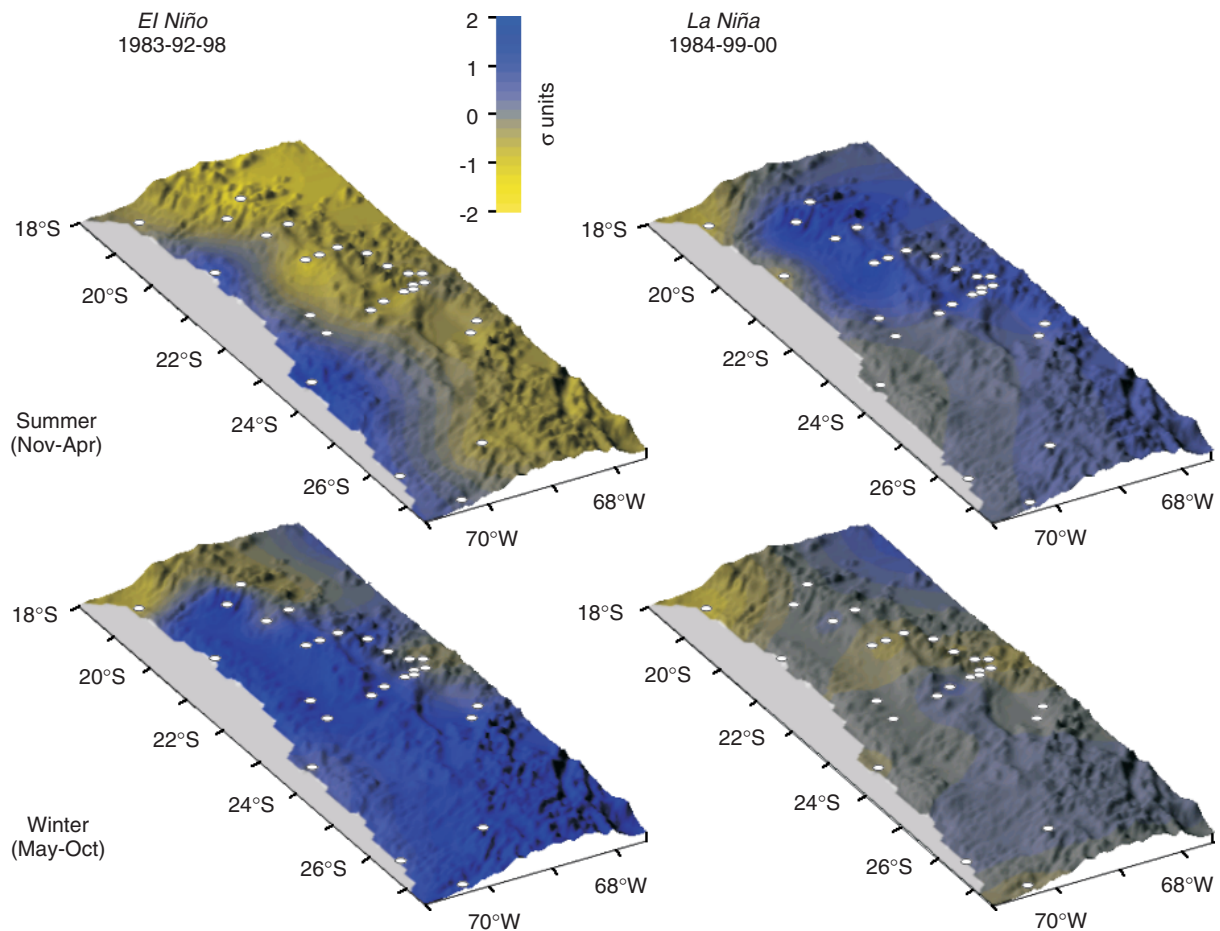


Figure 12. Precipitation anomalies associated with ENSO in the Atacama Desert. Mean values of summer and winter precipitations over three events for the stations shown are contoured using a kriging algorithm and overlaid on a digital elevation model. The coastal, southerly and winter distributions of precipitation associated with El Niño are clearly contrasted with the Andean and summer restricted precipitation associated with La Niña

Given the seasonal and spatial variation of precipitation extremes, it is essential to take these into account from a hydrological perspective. Figure 13 provides a schematic representation of the hydroclimatology of the Atacama Desert. Wet summers lead to flooding throughout the course of the major rivers including the coastal zones. Wet summers also lead to flows and flooding in the ephemeral rivers of the Andean slopes, as well as groundwater recharge either directly at higher elevations or via run-off infiltration at lower elevations. As previously shown, wet summers are largely linked to La Niña.

Wet winters, on the other hand, lead to snowfall in the southern central Andes and rainfall along the coast creating the well-known winter coastal floods (Garreaud and Rutllant, 1996) as well as flooding in the subsequent summer in the southern rivers (see also Waylen and Caviedes, 1990).

It is clear therefore that the hydrological response in the Atacama Desert is complex, and flooding and recharge may occur at the same place as a result of different mechanisms or at different places and different times due to the same mechanism. As a result, caution is required in assigning hydrological events and their associated sedimentary deposits to one or other of the ENSO phases and this may be a contributory factor to the contradictory results obtained from previous studies (e.g. Grosjean, 2001; Grosjean *et al.*, 2003; Rech *et al.*, 2003; Latorre *et al.*, 2004; Rech and Latorre, 2004). The hydrological complexity will be exacerbated over geologic time, as expansions and contractions in the Hadley circulation occur on centennial, millennial

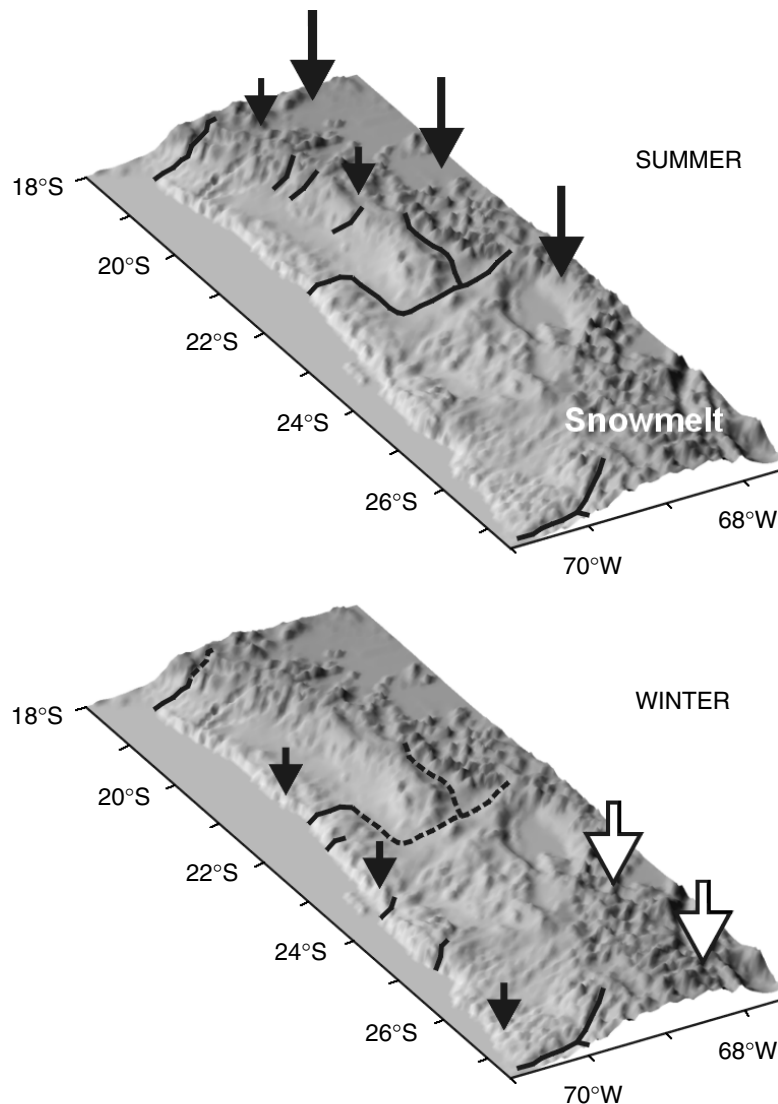


Figure 13. Schematic representation of the hydroclimatology of the Atacama Desert during wet years with recurrence intervals greater than six years. Enhanced summer rainfall is associated with La Niña, whereas coastal rainfall and Andean snowfall are associated with El Niño. Rivers maintained directly by precipitation are shown solid, while those maintained by groundwater drainage are shown dashed. Note the ephemeral rivers along the Pre-Cordillera in summer and along the coast in winter during years of enhanced precipitation

and orbital scales (e.g. Diaz and Bradley, 2004) causing the location of the summer/winter boundary zone to move south or north.

6.2. Hyper-aridity

The origin and causes of hyper-aridity, which carries the notion that there is no related fluvial activity, have been much discussed, as they have a bearing on the paleoclimatology of the Atacama Desert and its role in the origin of the Andes (e.g. Alpers and Brimhall, 1988; Montgomery *et al.*, 2001; Lamb and Davis, 2003; Hartley *et al.*, 2005). In discussing this it is essential to start with a clear definition: hyper-aridity is defined by UNEP (1997) as those zones having a ratio of mean annual precipitation to mean annual potential evaporation (MAE) of less than 0.05. This incorporates virtually the whole of the Atacama Desert

between 15° – 30° S, from sea level to 3500 m a.s.l. However, as previously noted and as clearly displayed in Figure 3, a zone of extreme hyper-aridity where MAR/MAE is less than 0.002 exists, associated with the boundary between summer and winter rainfall and the Central Valley where low rainfall is coupled with high evaporation.

By comparison, the Namib Desert of southwestern Africa, which occurs in the same zonal and western continental position, does not have a high coastal scarp and is backed by mountains that rarely exceed 2000 m in elevation. The Namib has generally higher rainfall than the Atacama (at 22° S coastal rainfall at Swakopmund is 23 mm year^{-1} , and at Windhoek, 524 m a.s.l., 250 km inland, rainfall is 363 mm year^{-1}) and the MAR/MAE ratio is never less than 0.02. Thus the Namib Desert may also be classified as hyper-arid, but it does not suffer from the same extremes as the Atacama Desert.

The Atacama and Namib Deserts are both associated with the descending limb of the Hadley circulation and have cold eastern ocean boundary currents offshore. The Namib, however, lacks a localized boundary layer cell decoupled from the ocean (the Rutllant cell of the Atacama), which might prevent the passage inland of limited Atlantic moisture, and the relatively low mountains inland allow the interchange of air masses with the interior so that no rain shadow develops (Tyson and Preston-Whyte, 2000). It might reasonably be inferred therefore that the hyper-aridity of both deserts is due to their zonal and western continental locations, but that the Rutllant cell and the rain shadow developed by the Andes create the extreme hyper-aridity of the Atacama Desert.

This has important implications for studies on Andean evolution, suggesting that the rain shadow that developed during the uplift of the Andes created positive feedback in the creation of extreme hyper-aridity. As a consequence, geological models that attribute the growth of the Andes to the onset of hyper-aridity as a result of Cenozoic climate change also need to take into account the impact of the rise of the Andes on atmospheric circulation.

While it is true that fluvial activity is very limited in hyper-arid zones, it still exists and does create geomorphological modifications. Most activity is dominated by incision and erosion rather than deposition, which is limited and localized (Rigsby *et al.*, 2003; Rech and Latorre, 2004). Thus the conditions that led to hyper-aridity are most likely to have developed during the period from 25–14 Ma (Alpers and Brimhall, 1988; Dunai *et al.*, 2005) but further intensified as the Andes rose above 2000 m around 10 Ma creating the rain shadow (Houston and Hartley, 2003; Evenstar *et al.*, 2005) and possibly the Rutllant cell.

7. CONCLUSIONS

Summer precipitation is largely restricted to the high-altitude part of the Atacama Desert with an easterly source, thus generating a rain shadow over the western Andean slopes. Extreme events tend to be associated with La Niña. By contrast, winter precipitation is rather more widely distributed, increasing towards the south because of its largely (westerly) frontal-system origin. At high elevations, winter precipitation is usually in the form of snow. Extreme events tend to be associated with El Niño. The frequency of summer and winter extreme rainfall is, however, rather less than ENSO events, pointing to synoptic controls (different for summer and winter, as described above), which are facilitated by ENSO rather than directly forced by it.

This variation of precipitation in space and time leads to a complex hydroclimatological system with various implications. Firstly, surface water floods occur in summer associated with La Niña throughout the central and northern Atacama Desert, but are caused by the summer melt of the previous winter's snow in the southern Atacama Desert and are associated with El Niño. Secondly, surface water floods occur in winter along the coastal Atacama Desert associated with El Niño. Thirdly, since the hydrologic systems are non-linear, flooding and recharge have a higher threshold for initiation, meaning they occur less frequently than precipitation extremes, but when they do, their impact is considerably magnified. Finally, as a result of the hydroclimatological complexity, which is likely to have been compounded by past changes in the extent and intensity of the Hadley circulation, paleoclimate, wetland and flood deposit studies cannot automatically assume wet or dry conditions associated with one specific

phase of ENSO. Furthermore, the frequency of extreme precipitation, typically decadal for events greater than 1σ and centennial for events greater than 2σ , renders present-day hydrological studies and water resource evaluations problematic, unless cognizance of the spatio-temporal variations is incorporated into them.

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